ads1-hw-examples (/github/BenLangmead/ads1-hw-examples/tree/master) / hw4_scss.ipynb (/github/BenLangmead/ads1-hw-examples/tree/master/hw4_scss.ipynb)

In [1]:
```
# This function is given in the assignment
from scs import scs

# You have to implement this function, or something like it
from scs_list import scs_list
```

Example 1

In [2]:
```
strings = ['ABC', 'BCA', 'CAB']
```

In [3]:
```
# Returns just one shortest superstring
scs(strings)
```

Out[3]: 'ABCAB'

In [4]:
```
# Returns list of all superstrings that are tied for shorest
scs_list(strings)
```

Out[4]: ['ABCAB', 'BCABC', 'CABCA']

Example 2

In [5]:
```
strings = ['GAT', 'TAG', 'TCG', 'TGC', 'AAT', 'ATA']
```

In [6]:
```
# Returns just one shortest superstring
scs(strings)
```

Out[6]: 'TCGATGCAATAG'

In [7]:
```
# Returns list of all superstrings that are tied for shorest
scs_list(strings)
```

Out[7]:
```
['AATAGATCGTGC',
 'AATAGATGCTCG',
 'AATAGTCGATGC',
 'AATCGATAGTGC',
 'AATGCTCGATAG',
 'TCGAATAGATGC',
 'TCGATAGAATGC',
 'TCGATGCAATAG',
 'TGCAATAGATCG',
 'TGCAATCGATAG']
```